Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 January 2026
Team ID	
Project Name	FORMULA 1 Race Intelligence Dashboard Analysis Report (2010-2026)
Maximum Marks	4 Marks

Data Flow Diagrams:

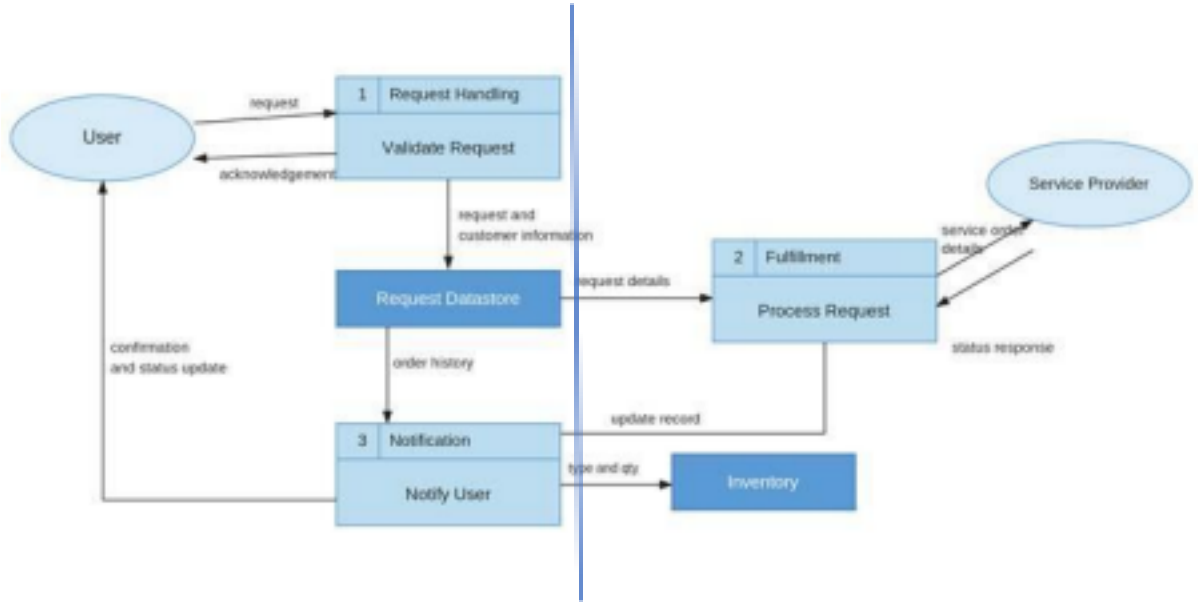
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).

Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Analyst	Data integration	US-1	As an analyst, I can connect the five workbook sheets to analyze related race data.	Inspect headers and define relationships.	Tables and join keys are documented; no accidental row multiplication.	High
Analyst	Data quality	US-2	As an analyst, I can identify missing values and invalid data types before visualization.	Check nulls, duplicates and numeric/date types.	Quality checks are recorded and ambiguous values are not silently imputed.	High
Strategist	Constructor comparison	US-3	As a strategist, I can compare constructor points by season.	Create line chart and team/season filters.	Season trend and selected teams update correctly.	High
Broadcast analyst	Driver leaderboard	US-4	As a broadcast analyst, I can rank drivers by points and wins/podiums.	Create sorted bar chart and tooltips.	Ranking matches source aggregations and responds to filters.	High
Race strategist	Tyre analysis	US-5	As a strategist, I can compare compound usage and lap times by tyre age.	Use lap_intelligence fields.	Compound and tyre-age views show available data only.	High
Technical analyst	Reliability	US-6	As a technical analyst, I can review DNF count and reason by team/season.	Build DNF charts and circuit/weather comparison.	DNF totals match race_results and reason nulls are disclosed.	High
Viewer	Story and export	US-7	As a viewer, I can follow a sequence of findings and inspect the filters.	Build Tableau Story from completed dashboards.	Story pages have descriptive	Medium

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
					captions and no placeholder findings.	
Customer Care Executive						
Administrator						

Data Flow Description

External entity: analyst/viewer. Data source: F1 Racing Project Dataset(1).xlsx. Process 1: validate and relate sheets. Process 2: calculate performance/reliability metrics. Process 3: render worksheets and dashboards. Process 4: publish story/export report. Data store: source workbook and Tableau workbook.

DFD text flow: Source workbook → validation & relationships → calculated fields → worksheets → dashboards → story/presentation. Feedback loop: analyst selects filters and drills down; the view updates without changing the source workbook.